mandate

Execution Evidence

Public transactions, recovery observations and verification limits

2026 FinTechathon Topic F: Agent Wallet

Mandate Labs

Nanyang Technological University, Singapore

Member	Role	Programme	Study year
Mengzhong Ma	Leader	Computational Finance	PhD, year 4
Sat Chidananda	Member	Computer Science	PhD, year 1
Vrushank Ajay Ahire	Member	Computer Science	PhD, year 1
Wilfried Ariel Mulyawan	Member	Computer Science	PhD, year 1

Mengzhong Ma	mengzhon001@e.ntu.edu.sg
Sat Chidananda	SAT001@e.ntu.edu.sg
Vrushank Ajay Ahire	vrushank001@e.ntu.edu.sg
Wilfried Ariel Mulyawan	wmulyawa001@e.ntu.edu.sg

Advisor: Bao Te, Nanyang Technological University.

Correspondence: Mengzhong Ma, team leader.

Member names, roles, programmes and study years reproduce the submitted registration record. The source registration files are not part of this public document.

Document revision: LaTeX revision 1, 20 October 2026.

Implementation reference: tested source commit

`29d10422954d36f0e37c2d89336986d10d014aba`

https://github.com/MandateLabs666/mandate-wallet

This is a testnet engineering document. Observed results, implementation assumptions and research extensions are identified separately. Preparing this document does not constitute an official competition submission.

Contents

1 Public Base Sepolia execution case **2**

1.1 Roles, account and scope 2

1.2 Transactions and separate outcomes 2

1.3 Independent evidence status and finality interpretation 3

2 How a reviewer should inspect the evidence **3**

3 Read-only verification and expected outcome **3**

4 Source identity, editing and fee interpretation **4**

5 Integrity and disclosure boundary **5**

1 Public Base Sepolia execution case

1.1 Roles, account and scope

The native run used chain 84532 with account `0xa794DAa86ff10920B75a4de3b0E752586e8703D2`. Its owner was `0xaA987fC286434D205669814Ba46b1d8df824fB36`, signing through Rabby. The asset was native Base Sepolia test USDC at `0x036CbD53842c5426634e7929541eC2318f3dCF7e`. The session signer was `0x818D3bd3ddf805FA9916AB50813d0EB67e40862f`, and the payee was `0x5B608Ea8219185dc5C971D2924913bF1e60fCEc1`. The owner key was not supplied to Claude or the production gateway.[3]

The root was `0x65a5386f35ae1dc848584bd51d1a8211110aa02b8c5264238aa4d4aa7d5a8c89`. The two-hour session was `0x0263e35063f897a6639f0272e8d6b8a3ff3d19c772f5d8b04baf05ae7c4ebd47`. These are historical identifiers, now subject to expiry and confirmed revocation. They are not reusable live access credentials.

1.2 Transactions and separate outcomes

The account received 150000 atomic USDC units, independently of native gas funding to the execution role. The successful task registered and paid 10000 atomic units once. Payment nonce `0xa65dbb50c51b68982cb7c19d440efad2bf574303856b0bb48a949f82f7cedddb` and settlement hash remained unchanged during paid-service recovery. Final observed wallet balance was 140000 units. The settlement and revocation rows below are the principal explorer evidence; all eight rows have exported receipts and logs.

Table 1: Eight key public transactions. Hashes are clickable Base Sepolia explorer links.

Purpose	Block	Transaction hash
account	48316879	0x2ba48a2f3197c6dfac2b3c5579b0e3e01350a2ba1290d019d044a685ed21cb26
fund gas	48318701	0x6ddf321b74a123e0ac7ee5007042c80dd667ac25a993f790622c258a91aebe2e
fund usdc	48318779	0x3bb4b38d34ddea128a261e8c5da4ca28dfd2369f8671f2b4a718ed8c25ff61e2
task	48324013	0x8336dc072584e2c29201de51e6d7ee1d44ba5c0ddf475188abcee7f0973b7505
session	48324110	0x6311ac19a4b212aa656d2e8b74a631f5c99fc0a109d3bfdf273c93ab142a1180
registration	48324434	0xdad95b5ada48d537a8fcd117be8ebab442d2032b4c1ac18810d5dd9d77d8b1e1
settlement	48324438	0xc23d943b69c94a5a8468ab32ee9b454ab6b57f7681ba4b93f16e4d58ae840c5d
revocation	48324929	0x5ba3feb1febde1fc2637c524ac031aa17e0a9f583bf39fc843bb865162d69a6e

The service outcome first recorded paid/failed/unchecked with HTTP 500, then paid/returned/-valid with HTTP 200. The valid result means the disclosed fixed sample matched the explicit deterministic checker. It does not establish an accurate market observation. After owner-signed session revocation, Claude attempted the unused BTC sample and received `AGENT_GRANT_INACTIVE`.

The step remained unregistered, no service was requested and no new transaction or USDC transfer occurred.

The public evidence retains an earlier expired, unsettled ETH order and an explicitly authorized new task in the same wallet. The wallet was not redeployed and no second top-up was hidden as recovery. An updated connector version 0.2.1 and its installed bundle fingerprints are also recorded. The raw run came from a working tree with changes; its file and bundle hashes, rather than Git HEAD alone, identify its source.

1.3 Independent evidence status and finality interpretation

The standalone verifier report confirms checkpoint integrity, request disclosure, ActionProof, registration authority, budget, canonical token payment, merchant offer/receipt and fixed-result checks. It has `requiredChecksPassed=true` and `overall=incomplete`. Independent HTTPS source evidence and complete history are missing. The workspace button lacked a trusted verification profile during recording. The successful report came from the standalone verifier with the explicit trust file, not from that button.

Two RPC providers reported all eight transactions below their common finalized height of 48325041 at the recorded observation. The exported receipts preserve block hashes and fee components. Provider agreement is a trust-based observation, not a locally verified consensus or rollup proof. The code and evidence are available for read-only review without a signer or new payment.

The picture edit contains actual UI captures with cuts, crops, scaling and held views. It is not an uninterrupted or synchronized multi-feed recording. The current picture duration is 290 seconds, with user narration pending in the material preparation snapshot. The merchant and facilitator remain controlled fixtures. No arbitrary Bazaar, mainnet, signed-installer or independent audit acceptance follows from this run.

2 How a reviewer should inspect the evidence

Start with `run.json` for chain, account, role addresses, source fingerprints and installed connector identity. Inspect `transactions.json` and `final-transactions.json` for the eight key receipts, logs and network-fee components. The export's `SHA256SUMS` covers the evidence files. An explorer screenshot is a useful presentation aid; the exported canonical transaction data carries the checkable financial facts.

`service-and-orders.json` separates first failed delivery from recovered success. `checks.json` summarizes payment amount, nonce/hash identity, final balance, independent result state and refusal. `refusal-evidence.json` records that the unused BTC step was rejected with no new registration, service request or transaction. A pre-broadcast refusal is not represented as a mined reverted transaction.

`purchase-bundle.json` contains disclosed signed checkpoints and ActionProof. `verifier-trust.json` independently pins the expected code, account, authority, request, merchant, budget and RPC inputs. `independent-verification.json` is the resulting report. The trust file must be reviewed separately from the bundle; a self-asserted wallet address is not a root of trust.[3, 4]

3 Read-only verification and expected outcome

Install the repository's frozen dependencies, then run from its root:

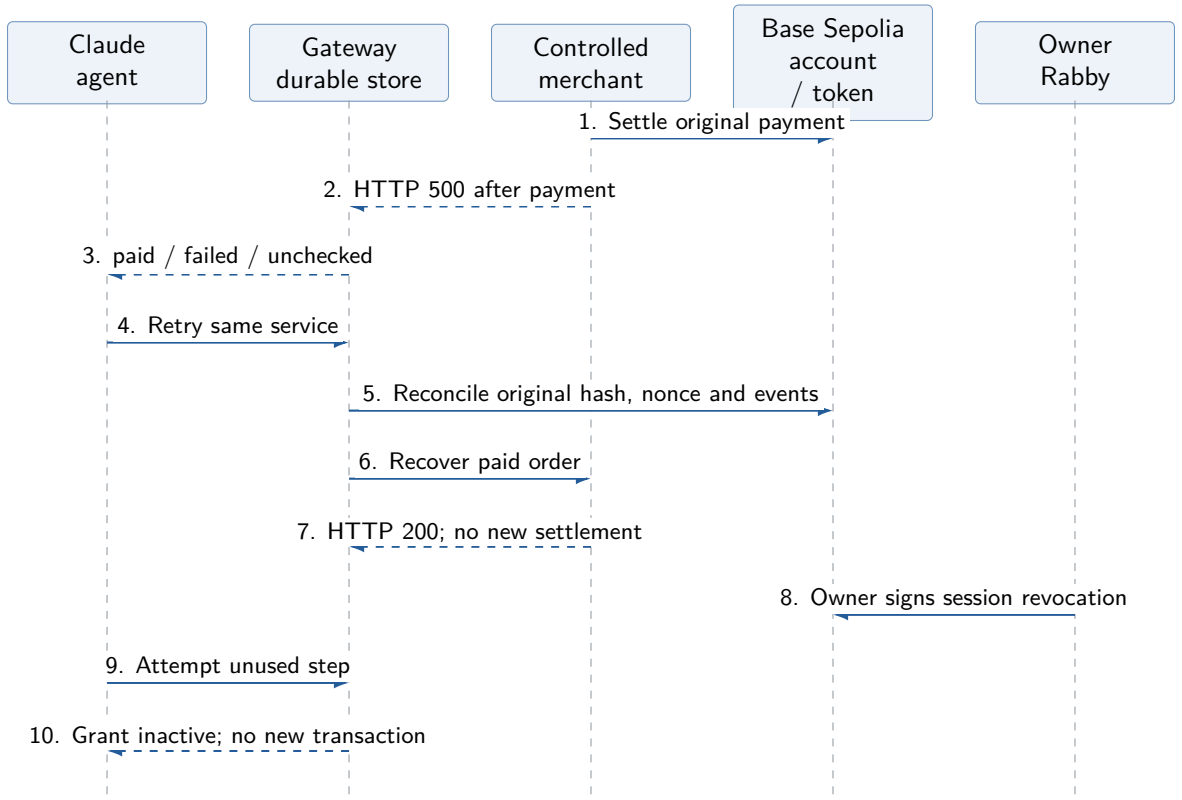


Figure 1: Observed delivery recovery and subsequent owner revocation. An expired, never-settled commitment requires a different handling path, as described in the text. The video preserves that earlier history separately.

```
node --import tsx mandate/evidence/cli.ts \
  --bundle competition/video/claude-desktop-base-v1/execution-evidence/purchase-bundle.json \
  --trust competition/video/claude-desktop-base-v1/execution-evidence/verifier-trust.json \
  --output /tmp/mandate-public-verification.json
```

This command requires historical RPC access but no owner key, session key or new transfer. It can fail or remain incomplete if a provider cannot supply the required historic state. No recovery should create a new payment merely to replace missing read-only evidence.

The recorded required-check set confirms integrity, ActionProof, registration authority, budget and payment. Other disclosed request/merchant/result checks also pass in the recorded report. The expected overall grade remains incomplete because no independent HTTPS witness or complete-history anchor is supplied. Contract signatures and transfer authorizations have different responsibilities, and the verifier checks the token events rather than treating a used nonce alone as paid.[1, 2]

4 Source identity, editing and fee interpretation

The actual recording source had uncommitted changes. `run.json` retains fingerprints for the exact account, gateway, transaction logic, verifier, onboarding and installed connector bundle. The later public code commit and its clean reproduction are different versioned artifacts. An accepted later build does not prove that the video originally ran from that clean commit.[5]

The video uses actual captured UI views with editorial cuts and held shots. `scene-map.json` links chapters to evidence; the adjacent video JSON maps shots to source-frame hashes. Debugging

and waiting were edited out. `recording-history.json` retains the earlier expired unpaid order, explicit owner-approved new task and connector update. The edit does not imply uninterrupted deployment-to-purchase execution.

Purchase cost is 0.01 test USDC. The exported network fees distinguish execution gas and L1 data fee in wei. Native gas funding is not a USDC service charge. The eight key transactions do not include every historical debugging transaction, so their sum must not be labeled the entire recording's total cost. The service/facilitator are controlled, the statistics are fixed and no market-price or third-party marketplace guarantee follows.

5 Integrity and disclosure boundary

The public export omits private keys, connection tokens, token payment-authorization signatures and raw signed transactions. It retains public evidence and merchant signatures needed for verification. Original registration attachments and the private proposal PDF are not included. The disclosed result is a public fixture. The independently graded limitations are part of the evidence, not reasons to relabel an incomplete report as confirmed.

References

[1] Francisco Giordano, Matt Condon, Philippe Castonguay, Amir Bandeali, Jorge Izquierdo, and Bertrand Masius. ERC-1271: Standard Signature Validation Method for Contracts. https://eips.ethereum.org/EIPS/eip-1271, 2018. Accessed 20 October 2026.

[2] Peter Jihoon Kim, Kevin Britz, and David Knott. ERC-3009: Transfer With Authorization. https://eips.ethereum.org/EIPS/eip-3009, 2020. Accessed 20 October 2026.

[3] Mandate Labs. Native Claude Desktop, Rabby and Base Sepolia execution evidence. https://github.com/MandateLabs666/mandate-wallet/tree/29d10422954d36f0e37c2d89336986d10d014aba/competition/video/claude-desktop-base-v1/execution-evidence, 2026. Recorded 19 October 2026; public evidence, trust configuration and source fingerprints.

[4] Mandate Labs. Mandate Wallet: tested source snapshot. https://github.com/MandateLabs666/mandate-wallet/tree/29d10422954d36f0e37c2d89336986d10d014aba, 2026. Source commit 29d10422. Accessed 20 October 2026.

[5] Mandate Labs. Clean-source reproduction and publication records. https://github.com/MandateLabs666/mandate-wallet/tree/a0156351bb2b0eca183c58b3071a3aae52cdbb94/artifacts/competition/source-delivery/submission20261020, 2026. 16 phases, 50 Solidity cases and 153 TypeScript tests; source commit 29d10422.